Rates & XVA

CVA, DVA and FVA of an Interest Rate Swap (Monte Carlo, Vasicek)

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1. Question

The textbook price of a derivative assumes that nobody defaults and that funding is free. In practice a bank adjusts this "clean" price for counterparty risk and funding costs. This project answers two questions on a 10-year interest rate swap:

- By how much does counterparty default risk change the value of the swap?
- How much of this adjustment disappears with collateral, and how much is added by wrong-way risk?

2. Definitions

Interest rate swap. Contract in which one party pays a fixed rate K and receives a floating rate on a notional N. The *payer* swap (pay fixed) gains value when rates rise.

Exposure. Amount lost if the counterparty defaults at time t: $V^+(t) = \max(V(t), 0)$. Its mean is the expected exposure $EE(t)$; its 95th percentile is the potential future exposure $PFE(t)$.

CVA (credit valuation adjustment). Expected loss due to the counterparty's default.

DVA (debit valuation adjustment). The mirror image: the gain linked to the bank's own default risk.

FVA (funding valuation adjustment). Cost of funding an uncollateralised position.

CSA threshold. Under a collateral agreement, the counterparty posts collateral as soon as the exposure exceeds a threshold H.

Wrong-way risk. Situation where exposure and default probability rise together.

3. Method

Step 1 – Interest rates. The short rate follows the Vasicek model, a mean-reverting Gaussian process:

$$dr_t = a\,(b - r_t)\,dt + \sigma\,dW_t, \qquad r_{t+\Delta} = r_t e^{-a\Delta} + b\,(1 - e^{-a\Delta}) + \sigma\sqrt{\tfrac{1-e^{-2a\Delta}}{2a}}\;Z.$$

The second expression is the exact transition, so the simulation has no discretisation bias. I simulate 40,000 paths (antithetic variates) on 240 time steps.

Step 2 – Swap revaluation. Vasicek gives the zero-coupon bond price in closed form, $P(t, t+\tau) = A(\tau)\,e^{-B(\tau)\,r_t}$ with $B(\tau) = (1 - e^{-a\tau})/a$. On each path and at each quarterly date t, the payer swap is therefore worth

$$V(t) = N\Big[\underbrace{1 - P(t, T_n)}_{\text{floating leg}} - \underbrace{K\,\delta \sum_{T_j > t} P(t, T_j)}_{\text{fixed leg}}\Big],$$

with no nested simulation. K is set to the par rate, so that $V(0) = 0$.

Step 3 – Adjustments. With $D(0,t)$ the stochastic discount factor, R the recovery rate and $PD(t) = 1 - e^{-\lambda t}$ the default probability implied by a CDS spread s through $\lambda = s/(1-R)$:

$$\text{CVA} = (1 - R_C) \sum_k \mathbb{E}\big[D(0,t_k)\,V^+(t_k)\big]\,\Delta PD_C(t_k),$$

$$\text{DVA} = (1 - R_B) \sum_k \mathbb{E}\big[D(0,t_k)\,V^-(t_k)\big]\,\Delta PD_B(t_k),$$

$$\text{FVA} = \sum_k \Big(s_{\text{borrow}}\,\mathbb{E}[D\,V^+] - s_{\text{lend}}\,\mathbb{E}[D\,V^-]\Big)(t_k)\,\Delta t.$$

Step 4 – Collateral. A CSA threshold H caps the exposure: V^+ is replaced by $\min(V^+, H)$.

Step 5 – Wrong-way risk. The default intensity becomes stochastic and linked to rates: $\lambda_t = \lambda_0 \exp(\beta z_t - \beta^2/2)$, where z_t is the standardised short rate. Its mean stays λ_0 for every β, so only the dependence changes, not the average credit quality.

4. Code

The core of the project is the revaluation of the swap on all paths at once (NumPy broadcasting), followed by the CVA sum. Full code: `xva_swap.py` (170 lines, NumPy and Matplotlib).

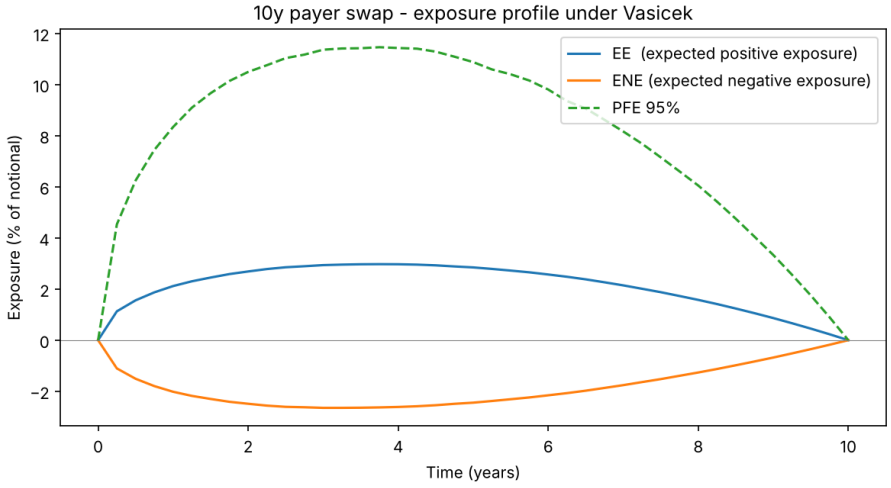
```
def swap_values(r_pay, pay_times, fixed_rate):
    """Payer swap value on each path at each payment date."""
    n = len(pay_times)
    values = np.zeros((r_pay.shape[0], n + 1))
    for k in range(n):
        t_k = 0.0 if k == 0 else pay_times[k - 1]
        tau = pay_times[k:] - t_k                          # remaining payments
        p = bond_price(r_pay[:, k][:, None], tau[None, :]) # paths x payments
        values[:, k] = NOTIONAL * ((1.0 - p[:, -1])
                                   - fixed_rate * DELTA * p.sum(axis=1))
    return values

pos = np.maximum(v, 0.0)                                   # positive exposure
cva = (1 - REC_C) * np.sum((disc * pos).mean(axis=0)[1:]
                           * default_increments(SPREAD_C, REC_C, grid))
```

5. Results

Parameters (illustrative, not calibrated to market data): $a = 0.10$, $b = 3\%$, $\sigma = 1\%$, $r_0 = 2.5\%$; counterparty CDS spread 150 bps, bank spread 60 bps, recovery 40%; funding spreads 50 and 20 bps.

Validation. The Monte Carlo price of the 10-year zero-coupon bond is 0.77114 against 0.77106 in closed form (error 0.010%). The par swap rate is 2.6062%.



The exposure has the classical hump shape: uncertainty on rates grows with time, while the number of remaining cash flows falls. The expected exposure peaks near 3% of notional around year 4 and the 95% PFE near 11.5%. EE is larger than ENE because the curve is upward sloping ($r_0 < b$), so the payer swap tends to gain value.

Adjustment	bps of notional
CVA (Monte Carlo standard error 0.07)	24.57
DVA	10.03
FVA	5.58
Total adjustment $-\text{CVA} + \text{DVA} - \text{FVA}$	-20.12

Collateral			Wrong-way risk		
CSA threshold H	CVA (bps)	Change	β	CVA (bps)	Change
none	24.57	–	-0.5 (right-way)	13.38	-45.5%
5.0% of notional	18.28	-25.6%	0.0 (independent)	24.57	–
2.0%	9.72	-60.4%	$+0.5$ (wrong-way)	38.79	$+57.9\%$
1.0%	5.36	-78.2%	$+1.0$ (wrong-way)	52.31	$+112.9\%$
0.5%	2.82	-88.5%			

Reading.

- Counterparty risk costs about 25 bps of notional on this swap; the bank's own credit and funding bring the total adjustment to -20 bps.
- Collateral is the main lever: a threshold at 1% of notional removes 78% of the CVA.
- If the counterparty's credit worsens when rates rise ($\beta > 0$), exposure and default probability move together and the CVA can double. The dependence matters as much as the level of the spread.

6. Limitations and extensions

- One-factor Vasicek with constant parameters: it does not fit a market curve and allows negative rates. Hull–White (time-dependent b) would fit the initial curve exactly.
- Collateral is an instantaneous cap. A realistic model adds a margin period of risk (about 10 days), during which the exposure can still move.
- CVA and DVA are computed separately, without the first-to-default effect.
- Single swap. The natural extension is a netting set of several swaps, to measure the netting benefit.